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RECIRCULATING CRUSHING PLANT

Abstract

Systems, methods and apparatus are provided for crushing aggregate material. In some embodiments, in an operating configuration, a recirculating conveyor deposits material previously crushed by a crusher into the feed inlet of the crusher. In some embodiments, the recirculating conveyor is pivoted inward about a vertical axis into the operating configuration.

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Background/Summary

BACKGROUND

[0001] Crushing plants include crushers for crushing material such as aggregate material. Some crushing plants include recirculating conveyors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a side elevation view of an embodiment of a recirculating crushing plant.

[0003] FIG. 2 is a top view of the recirculating crushing plant of FIG. 1.

[0004] FIG. 3 is a side elevation view of another embodiment of a recirculating crushing plant.

[0005] FIG. 4 is a top view of the recirculating crushing plant of FIG. 3.

DESCRIPTION

[0006] Referring to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIGS. 1 and 2 illustrate a recirculating crushing plant **100** including a chassis **110** which is optionally portable or mobile (e.g., on wheels **115**, powered tracks, non-powered tracks, etc.). A vibratory feeder **210** (e.g., a grizzly feeder having an array of grizzly bars **220**) is optionally supported on the chassis **110**. The feeder **210** optionally receives a feed of material onto a region R.sub.1. Vibration of the feeder **210** optionally moves the feed of material to a region R.sub.2 where the material is optionally classified, with undersized material falling onto a cross-conveyor **280** for conveying away from the plant **100**. The feeder **210** optionally deposits oversized material (either directly or via an intermediate conveyor or feeder) into a crusher such as a horizontal shaft impact crusher **300** (or in other embodiments another crusher such as a vertical shaft impact crusher, cone crusher, jaw crusher, etc.). The crushed output of crusher **300** is optionally deposited (either directly or via an intermediate conveyor or feeder) onto a conveyor **120**.

[0007] Conveyor **120** optionally deposits material onto a vibratory classifier **400** (e.g., multi-deck incline screen, multi-deck horizontal screen, etc.). A conveyor **130** is optionally disposed to receive undersize material passing through the classifier **400** for conveyance away from the plant **100**. A cross-conveyor **140** is optionally disposed to convey material onto a recirculating conveyor **150**.

[0008] Recirculating conveyor **150** optionally has a transport configuration in which the length of the conveyor **150** is generally aligned with the chassis **110** and/or the travel direction of the chassis **110**. Recirculating conveyor **150** optionally has an operating configuration pivoted inboard (generally about the direction DR shown in FIG. 2) such that a head end H of conveyor **150** is disposed to deposit material onto region R.sub.1 or R.sub.2 of the vibratory feeder **200**. In some embodiments, the conveyor **150** deposits material generally in the transverse center of the feed of material on vibratory feeder **200**.

[0009] In some embodiments, conveyor **150** is at least partially supported on a distal end of a pivot arm **112** for pivoting about a generally vertical axis B. A proximal end of pivot arm **112** is optionally pivotally supported on chassis **110** for pivoting about a generally vertical axis A. The conveyor **150** is optionally supported on a support **114** (e.g., pivotal support) which also optionally permits pivoting about a vertical axis.

[0010] Pivoting of the pivot arm **112** and/or conveyor **150** permits inboard pivoting of the conveyor **150** about the direction DR. One or more actuators (not shown) are optionally disposed to pivot the pivot arm **112** relative to chassis **110** and/or to pivot the conveyor **150** (e.g., by pushing a generally forward portion of the conveyor **150** in an inboard direction).

[0011] The conveyor **150** optionally has a rearward portion **152** and forward portion **154** joined at a

horizontal pivot **155**. An actuator **157** is optionally disposed to raise and lower the forward portion **154** (e.g., for operation and transport, respectively, and/or for positioning of head end of conveyor **150** relative to crusher **300**).

[0012] Referring to FIGS. **3** and **4**, another embodiment of a plant **100'** is illustrated having a modified conveyor **150'** configured (e.g., sized) to deposit material (e.g., directly) into a feed inlet **302** of the crusher **300** in the operational configuration of the conveyor **150'**. In some embodiments, the conveyor **150'** includes a chute **159** or other intermediate structure disposed at a head end of the conveyor **150** and configured to guide material to the inlet **302**. In some embodiments, the conveyor **150'** is configured to deposit material generally at a lateral center of the feed inlet **302** in the operational configuration of the conveyor **150'**. Material entering the lateral center of the feed inlet **302** is optionally laterally centered on a blow bar rotor of the crusher **300**.

[0013] In some embodiments, an intermediate structure such as a slide, chute, etc. is disposed under and optionally attached supported by the head end of conveyor **150** (or conveyor **150'**). The intermediate structure is optionally configured to receive at a first deposition location from the conveyor **150**, slidably support material deposited by the conveyor **150** as the material moves downward by gravity and inboard toward a second deposition location at which the material falls from the intermediate structure. The second deposition location is optionally aligned with the lateral center of the feed inlet of the crusher along a vertical plane extending in the travel direction of the plant **100**. In various embodiments, the second deposition location is located above the crusher inlet, above the feeder, etc.

[0014] Although various embodiments have been described above, the details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art. Accordingly, the scope of the present disclosure is intended to be interpreted broadly and to include all variations and modifications within the scope and spirit of the appended claims and their equivalents. For example, any feature described for one embodiment may be used in any other embodiment.

Claims

1. A portable recirculating crushing plant, comprising: a portable chassis; a feeder supported on said chassis, said feeder having a discharge end; a crusher supported on said chassis, said crusher having an upward-facing feed inlet in an upper portion thereof to receive material deposited vertically therein, said feed inlet being separate and spaced apart from said discharge end, said discharge end of said feeder being positioned to deposit material into said feed inlet of said crusher; a vibratory classifier supported on said chassis; a first conveyor disposed to convey material from said crusher to said vibratory classifier; a second conveyor disposed to receive material from said vibratory classifier, said second conveyor having a head end and a tail end, said second conveyor having a length, said second conveyor having a chute supported thereon, said chute being disposed to receive material deposited from said head end of said second conveyor, said chute having a chute outlet, said second conveyor having a transport configuration and an operating configuration, said second conveyor being pivotable about a pivot axis B, said pivot axis B being generally parallel to a pivot axis A, said pivot axis B being spaced apart from said pivot axis A, whereby said second conveyor is pivotable about pivot axis A and pivot axis B, said second conveyor being pivotable between said transport configuration and said operating configuration, wherein in said operating configuration, said second conveyor is oriented to convey material in an at least partially inboard direction toward a longitudinal centerline of said chassis, wherein in said operating configuration, said length of said second conveyor is such that said chute is disposed to guide material directly into said feed inlet without contacting said feeder, and wherein in said transport configuration, said head end is disposed outboard of said crusher, wherein said crusher is disposed such that second conveyor is able to pivot into said operating configuration with said head end above said crusher

without interfering with said crusher.

2. The portable recirculating crushing plant of claim 1, wherein said crusher comprises a blow bar rotor, wherein in said operating configuration said second conveyor is disposed to deposit material onto a lateral center of said blow bar rotor along an axis of rotation of said blow bar rotor.

3. The portable recirculating crushing plant of claim 1, wherein said second conveyor comprises a chute, said chute being disposed to deposit the material into said feed inlet.

4. The portable recirculating crushing plant of claim 1, further comprising: a vibrating feeder supported on said chassis, wherein said crusher is disposed to receive material from said vibrating feeder.

5. The portable recirculating crushing plant of claim 1, wherein said second conveyor is configured to deposit material at a lateral center of said feed inlet along a lateral direction normal to a longitudinal axis of said chassis in the operational configuration of said second conveyor with said tail end being offset from said lateral center.

6. The portable recirculating crushing plant of claim 1, wherein said crusher comprises a blow bar rotor, wherein said second conveyor is configured to deposit material at a lateral center of said blow bar rotor with said tail end being offset from said lateral center.

7. The portable recirculating crushing plant of claim 1, further comprising a horizontal pivot along said second conveyor between said head end and said tail end, said horizontal pivot pivotally supporting said second conveyor and constraining said second conveyor to pivot about a third horizontal axis.

8. A method of crushing material, comprising: supporting a pivot arm on a chassis for pivoting about a first axis A; supporting a return conveyor on said pivot arm for pivoting said return conveyor about a second axis B; repositioning a tail end of said return conveyor from a relatively inboard transport position to a relatively outboard position by pivoting said return conveyor about said first axis A and said second axis B; pivoting a head end of said return conveyor to an operational configuration in which said head end of said return conveyor is at least partially disposed above a crusher; depositing a feed of material from a discharge end of a vibratory feeder; with said crusher, said crusher having a feed inlet separate from said vibratory feeder, receiving said feed of material in said feed inlet and crushing material from said feed of material; conveying material from said crusher to a vibratory classifier; with said vibratory classifier, classifying material from said crusher; by said return conveyor in said operational configuration, conveying a subset of classified material in an at least partially inboard direction; and by said return conveyor in said operational configuration, conveying material along a return conveyor length and depositing said subset of classified material directly into said feed inlet of said crusher without said subset of classified material contacting said vibratory feeder, wherein said return conveyor length is configured such that in said operational configuration, at least one of a head end or chute of said return conveyor is disposed directly above said feed inlet and not directly above said vibratory feeder.

9. The method of claim 8, wherein crushing material comprises impacting material against a plurality of surfaces by a blow bar rotor.

10. The method of claim 9, wherein conveying material from said crusher to a vibratory classifier comprises centering said feed of material on said blow bar rotor with said tail end being off-center from said blow bar rotor.

11. The method of claim 8, further comprising: constraining said return conveyor to pivot about a third pivot axis disposed between said tail end and said head end of said return conveyor.

12. A method of crushing material, comprising: supporting a return conveyor for pivoting said return conveyor about a first axis A and a second axis B parallel to and spaced apart from said first axis A; repositioning a tail end of said return conveyor from a relatively inboard transport configuration to a relatively outboard position by pivoting said return conveyor about said first axis A and said second axis B; pivoting a head end of said return conveyor to an operational

configuration in which said head end of said return conveyor is at least partially disposed above a crusher, said pivoting being carried out without interfering with said crusher; depositing a feed of material from a discharge end of a vibratory feeder; with said crusher, said crusher having a feed inlet separate from said vibratory feeder, receiving said feed of material in said feed inlet and crushing material from said feed of material; conveying material from said crusher to a vibratory classifier; classifying material from said crusher; by said return conveyor in said operational configuration, conveying a subset of classified material in an at least partially inboard direction; and by said return conveyor in said operational configuration, conveying material along a return conveyor length and depositing said subset of classified material directly into said feed inlet of said crusher without said subset of classified material contacting said vibratory feeder, wherein said return conveyor length is configured such that in said operational configuration, at least one of a head end or chute of said return conveyor is disposed directly above said feed inlet and not directly above said vibratory feeder.

13. The method of claim 12, wherein crushing material comprises impacting material against a plurality of surfaces by a blow bar rotor.

14. The method of claim 13, wherein conveying material from said crusher to a vibratory classifier comprises centering said feed of material on said blow bar rotor with said tail end being off-center from said blow bar rotor.

15. The method of claim 12, further comprising: constraining said return conveyor to pivot about a third pivot axis disposed between said tail end and said head end of said return conveyor.
